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-THE CELL SYSTEM-

Animal Cell Diagram

Color the organelles according to the key.

cytoplasm (yellow) / nucleus (gray) / endoplasmic reticulum (purple)
/ mitochondria (red) / vacuole (light blue) / lysosome (blue) /
Golgi apparatus (orange)

Hint: lysosomes are oval and vacuoles are rounded.

Which two cell structures are you not coloring?

Early finishers: work on test corrections (look at your test on schoology), science fair stuff, lesson 2 check

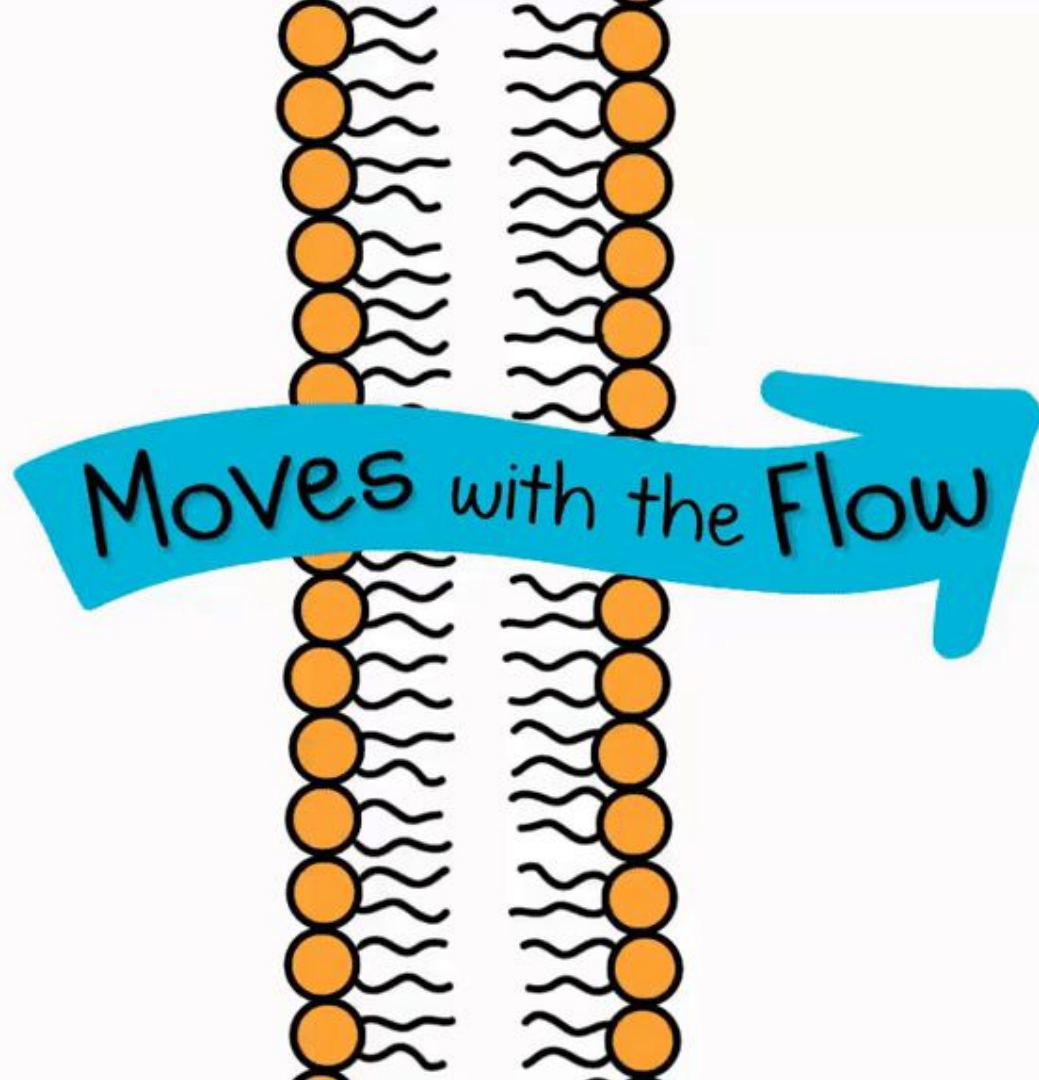
October 2, 2025

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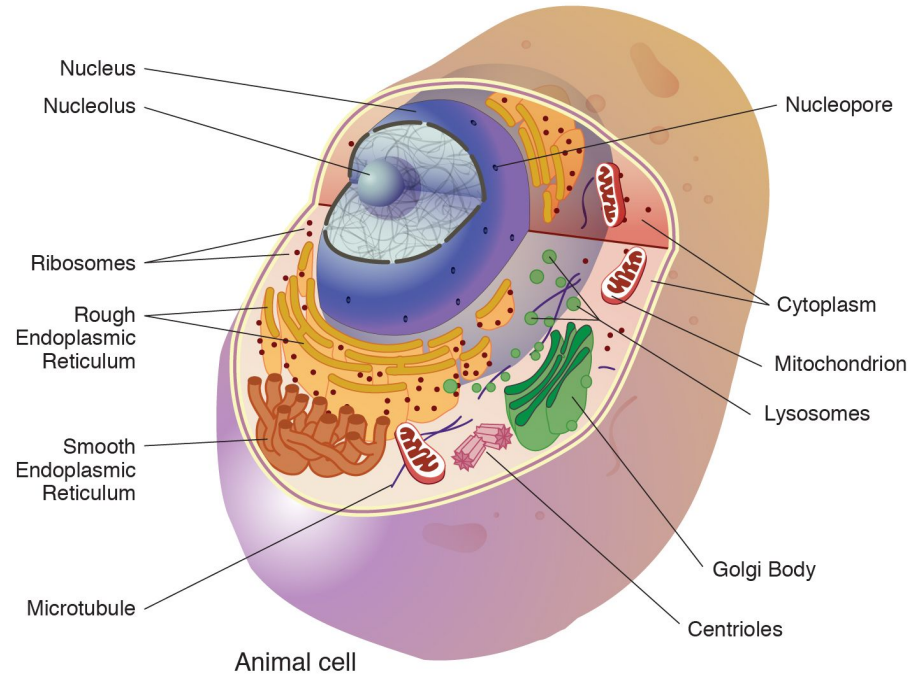
-THE CELL SYSTEM-

- Animal Cell Diagram

- L3 Obtaining and Removing

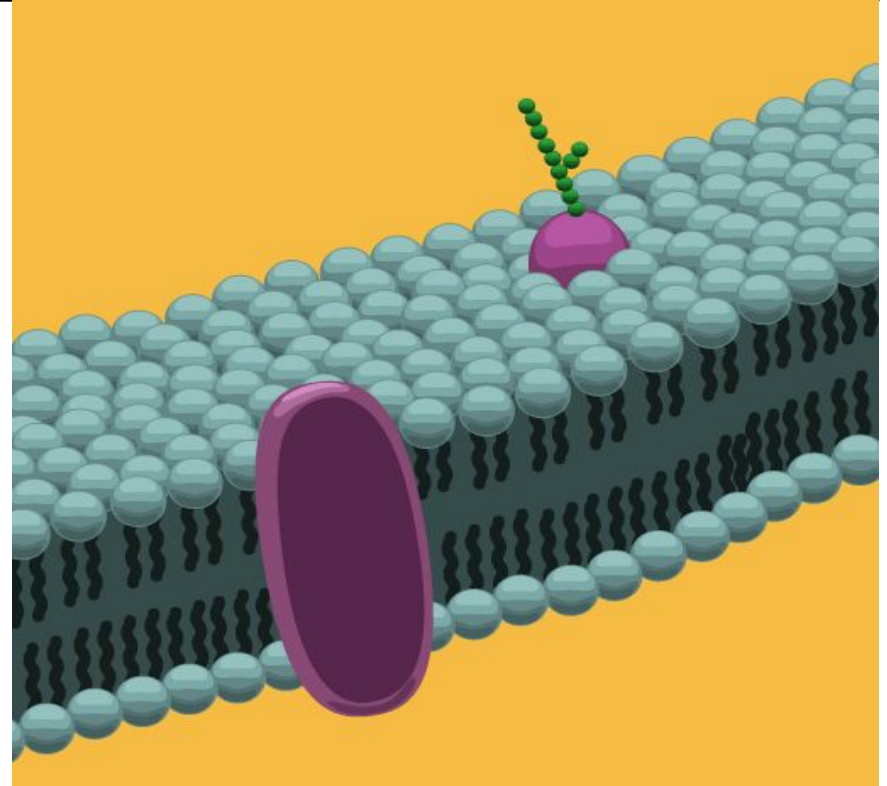


What do you know about the cell membrane?



Characteristics of the cell membrane

- Lets substances in and out
- It's flexible
- It's permeable
- Selectively permeable

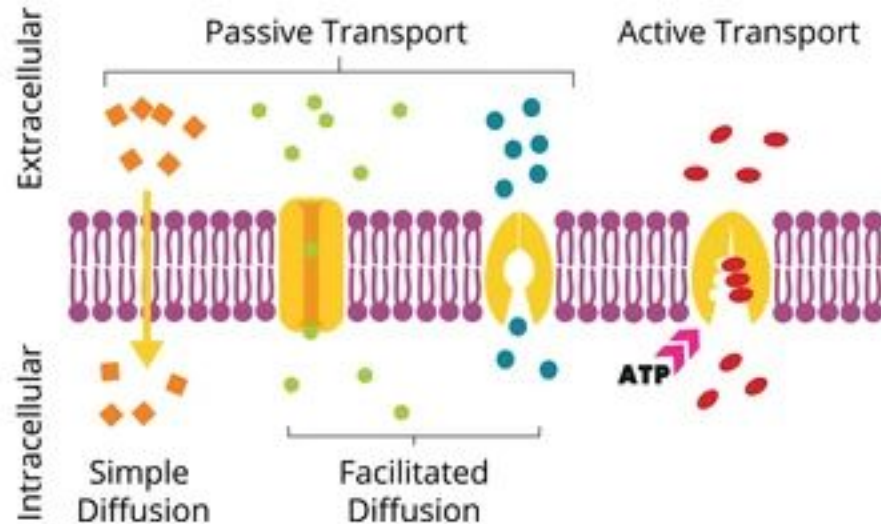


Substances move in and out of cells by two processes

Cell Transport

Passive Transport

Active Transport

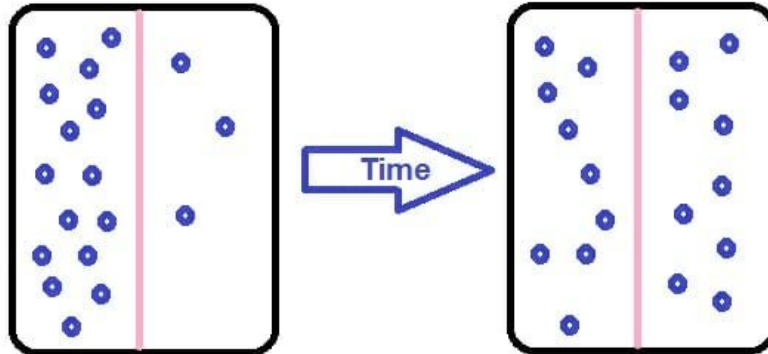


Passive Transport

- Requires no energy
- 3 types:
 1. Diffusion
 2. Osmosis
 3. Facilitated diffusion
- flow goes from high to low concentration

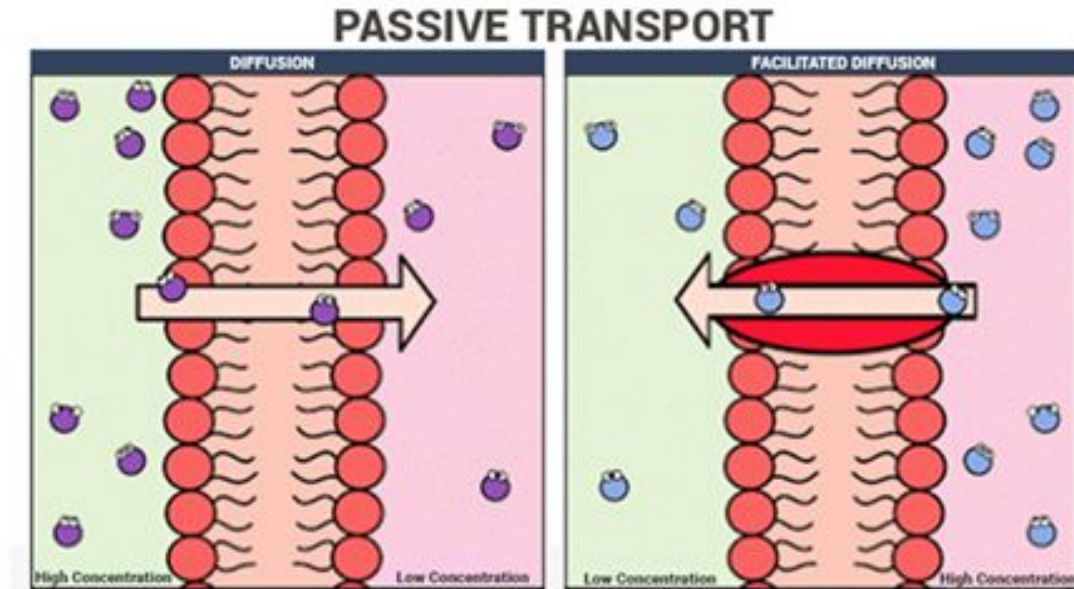
Diffusion

Movement of **particles** from areas of high concentration to areas of low concentrations. Ex.: CO₂, O₂



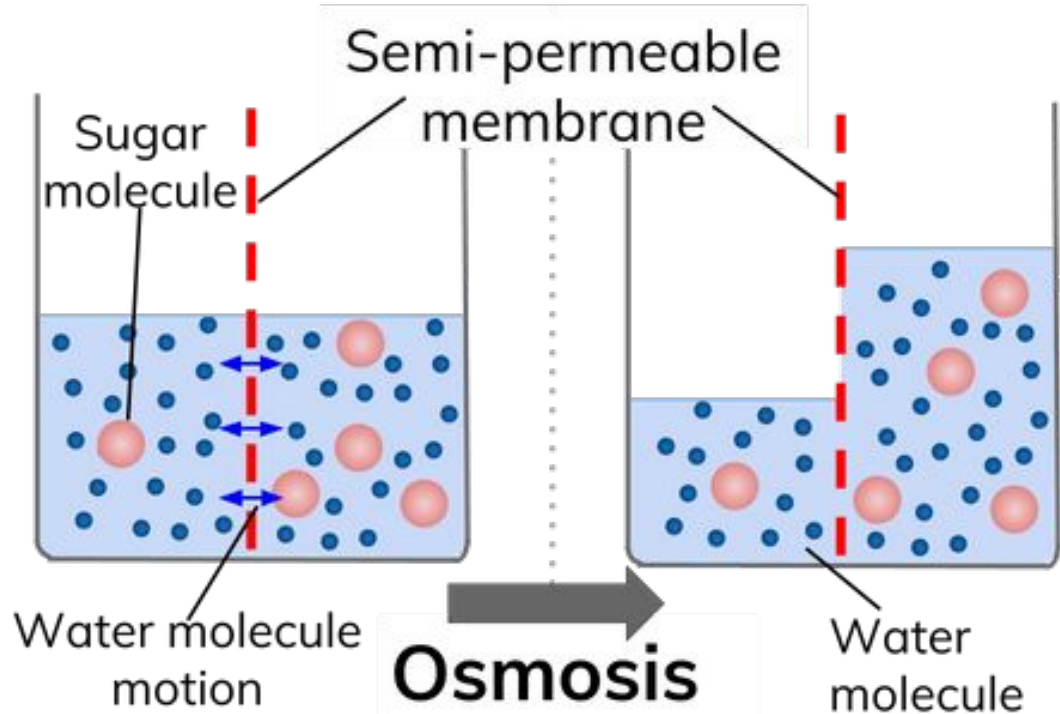
Facilitated Diffusion

Sugar cannot easily cross the cell membrane. Proteins in the cell membrane form channels through which the sugars can pass.



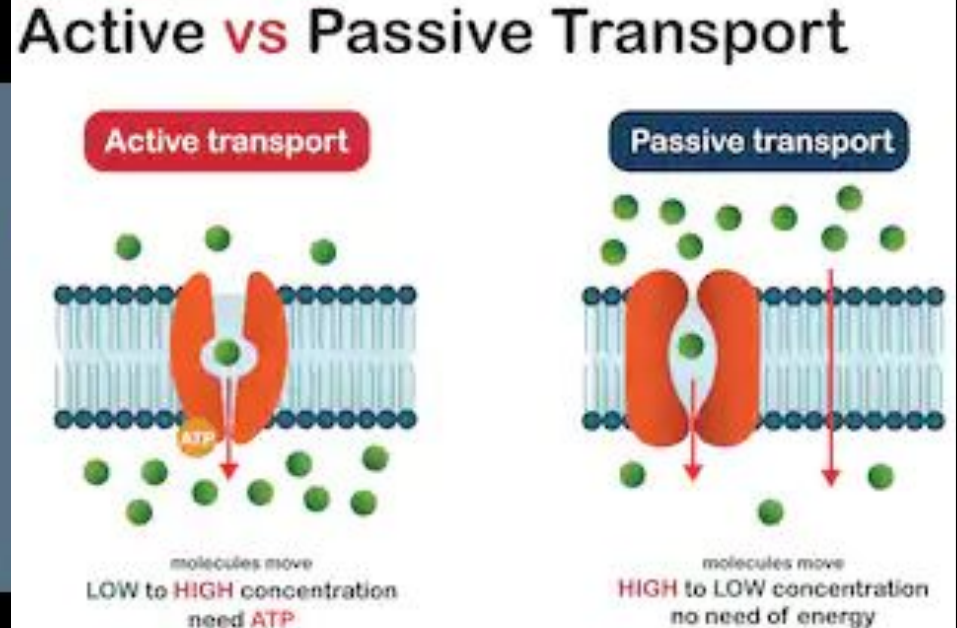
Osmosis

Diffusion of **water** molecules from a higher water concentration to lower water concentration through a semipermeable membrane.

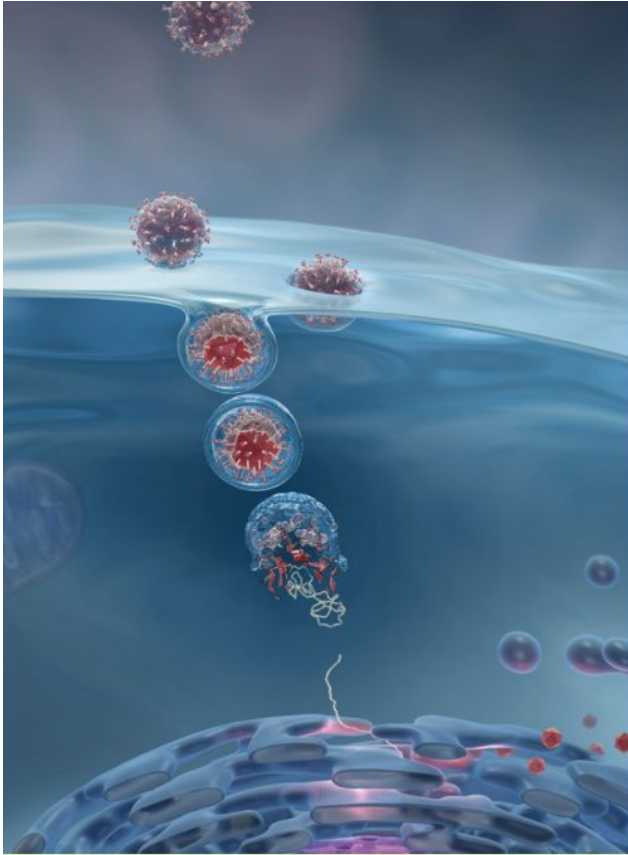


Active Transport

The movement of materials across the membrane from areas of low concentration to areas of high concentration.



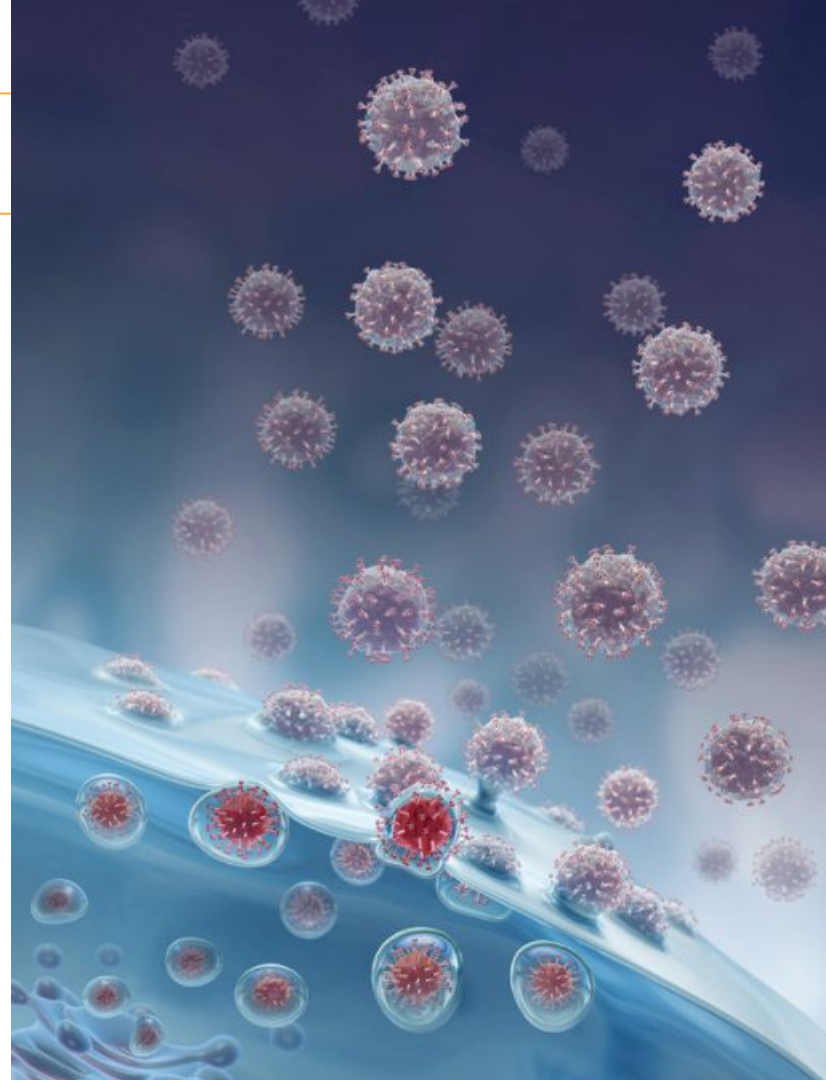
Moving Large Particles



- **Endocytosis**: the cell membrane takes larger particles (such as food particles) into the cell by changing shape and engulfing the particles.

Moving Large Particles

- Exocytosis allow large particles to leave the cell



October 6, 2025

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- Topic 2 Study Guide

Directions: Describe the function of each cell structure and indicate whether it is found in plant cells, animal cells, or both.

Structures:

- Cell Wall
- Cell Membrane
- Cytoplasm
- Nucleus
- Endoplasmic Reticulum
- Ribosome
- Golgi Apparatus
- Mitochondria
- Chloroplast
- Vacuole
- Lysosome

EARLY FINISHERS: page 84+85 - topic 2 review

Cell Wall

Protects and
supports the cell

PLANT

Cell Membrane

Forms
semi-permeable
barrier, controls
what goes in and
out of the cell
BOTH

Cytoplasm

Fills the cell, allows
movement inside

BOTH

Nucleus

Contains genetic
info, controls cell
activities

BOTH

Endoplasmic Reticulum

Helps make
proteins. overrated

BOTH

Ribosomes

ASSEMBLES
PROTEINS.
HONESTLY MVP OF
THE CELL.

BOTH

Golgi Apparatus

Packages stuff from the ER. overrated.

BOTH

Mitochondria

Converts food to energy the cell can use (ex. Active transport)

BOTH

Chloroplast

P H O T O
S Y N T H E S I S

PLANT

Vacuole

Cell storage. Food, water, waste, etc.

BOTH

Lysosome

Breaks down molecules.

ANIMAL

What's an analogy anyway???

A comparison of two things that are similar in important ways

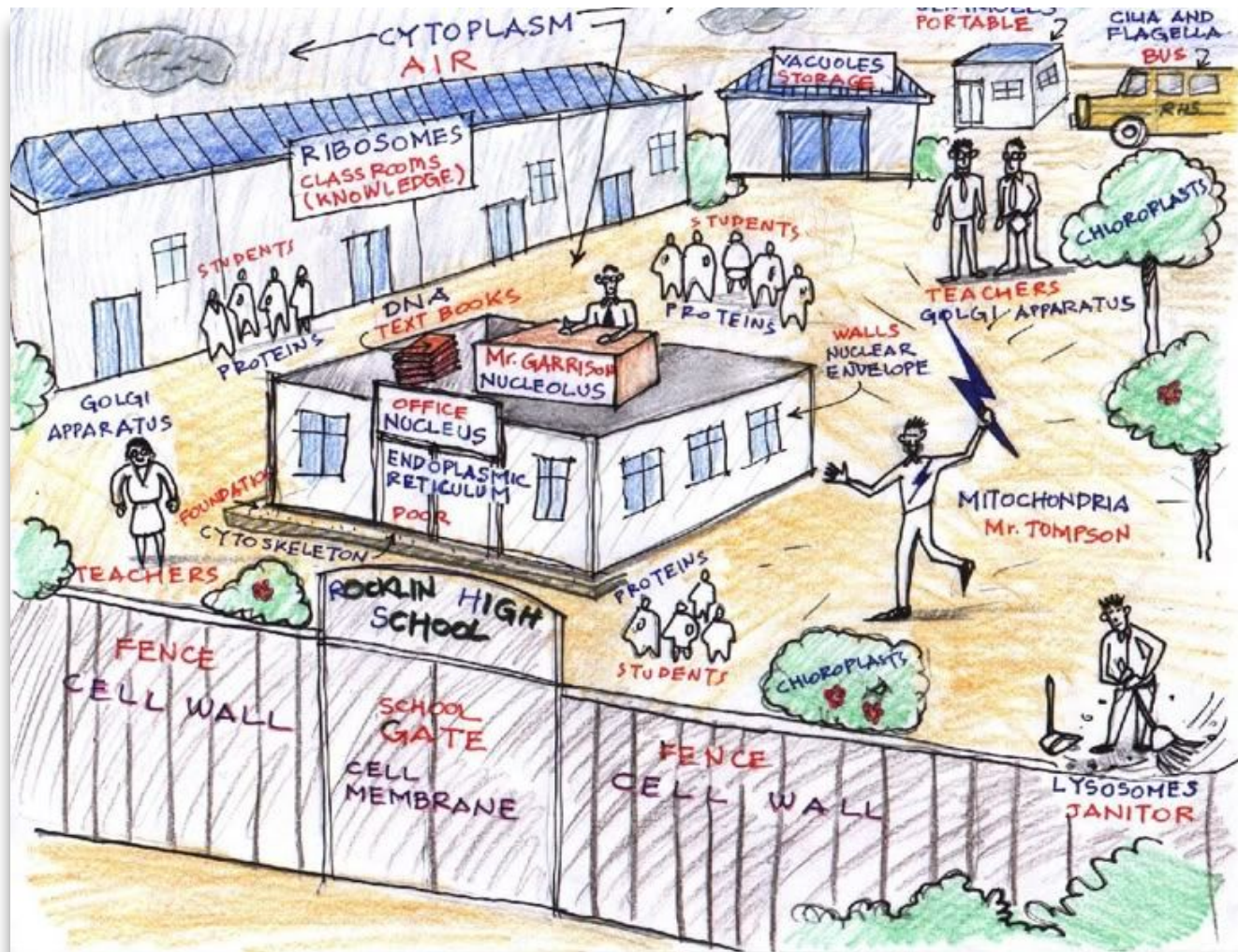
- Mx. Martin has used their classroom as an analogy for a cell!

Word: Analogous (just the adjective version of 'analogy')

- A human arm is analogous to a bird's wing.
- The corrosives cabinet is analogous to a lysosome.

You will be creating a Cell Analogy to demonstrate that you know how the different parts of a cell work! Also so that Mx. Martin can put something on the pink wall.

August 68, 2025



Today's Goal:

With your partner, complete the Cell Analogy Worksheet and begin work on your poster. The next block period will be for completing the cell poster.

THIS IS DUE monday?????

Today's Goal:

Finish the poster and the key? This will be the last dedicated class time you get for this.

You can listen to music while you work.

I have approximately five million of these blank papers so you can get a fresh one to start over anytime.

Due Date: next Friday.

Cell Theory:

- cells are the basic unit of life
- cells can only come from cells
- yep

Cell Transport:

Are the following passive or active transport?

What material might get moved with each of the following?

1. Diffusion - oxygen, carbon dioxide
2. Osmosis - WATER AND ONLY WATER
3. Facilitated diffusion - glucose
4. Active transport - low to high (get more glucose!) (pump out waste)
5. Endocytosis or Exocytosis - large sugars (cellulose), large proteins, white blood cells with viruses or bacteria

Go with the flow or against it? What's being moved? Big or small?

